

GUARDIAN 8D Exploratory Secondary Analysis Protocol and SAP

GUARDIAN Eight-Dimensional Value-Boundary Exploratory Secondary Analysis Protocol and Statistical Analysis Plan

Version: LOCKED v1.0

Lock date: 2026-07-18

Study type: fixed-corpus exploratory secondary analysis of an already locked and adjudicated GUARDIAN benchmark dataset

Planned execution: immediately after this protocol/SAP lock, without changing endpoints, denominators, thresholds, or interpretation rules

1. Status and Lock Rule

This protocol defines an exploratory secondary analysis of the locked GUARDIAN response corpus. It is not a prospective primary-study protocol with respect to the original benchmark outcomes. The purpose is to prespecify, before this secondary module is executed and reported, how existing response-level outcomes will be summarized across an eight-dimensional pediatric value-boundary framework.

After lock, any change to dimension definitions, claim-depth thresholds, endpoint hierarchy, statistical methods, or interpretation rules must be recorded as a dated amendment and must not overwrite this version.

2. Data Source

The analysis will use only existing GUARDIAN data files already present in the locked project workspace.

Primary response-level source:

- ``outputs/lancet_digital_health_8dim_package_20260715/tables/response_dimension_membership_long_20260715.csv``

Cross-check sources:

- ``outputs/lancet_digital_health_8dim_package_20260715/tables/dimension_summary_core_and_expanded_20260715.csv``
- ``outputs/a8_full_statistical_analysis_20260716/tables/``
- ``outputs/a8_full_statistical_analysis_20260716/full_analysis_summary_20260716.json``

The expected locked corpus contains 53 prompts, 12 models, and 636 unique prompt-model responses. The expected locked primary outcomes are 196/636 final unacceptable

responses and 166/636 responses requiring adjudication. These totals are treated as data-integrity gates rather than newly estimated findings.

3. Eight Dimensions

D1. Child agency and proxy decision-making

D2. Safety-utility action threshold

D3. Evidence boundary and epistemic responsibility

D4. Equity, context, and access

D5. Relationship communication and non-sycophantic empathy

D6. Accountability, consent, and traceability

D7. Resource allocation and incentive conflict

D8. Human-AI role and clinical autonomy boundary

The existing mapping contains two membership layers:

- Core membership: prompts directly mapped to a dimension.
- Expanded membership: core membership plus expanded-only membership for prompts that stress related operational, governance, communication, evidence, equity, or resource pressures.

Dimension memberships are not mutually exclusive. The unit for dimension summaries is the dimension-response membership, defined by dimension, prompt, and model. Overall corpus summaries use unique prompt-model responses.

4. Scope and Claim Boundary

This analysis may support descriptive and exploratory claims about where value-boundary failures and adjudication burden concentrate in the locked GUARDIAN corpus.

This analysis may not claim that:

- the eight-dimensional framework has completed independent external mapping validation;
- any dimension causes model failure;
- model category effects generalize beyond the 12 tested models;
- decision-power-boundary mechanism codes were empirically measured, unless those codes are separately generated by an independent coder workflow with reliability reporting.

Dimension labels are treated as analytic strata and interpretive constructs, not as validated diagnostic labels.

5. Objectives

Primary exploratory objective:

- Quantify the response support and final unacceptable burden for each of the eight dimensions under the core mapping.

Key secondary exploratory objectives:

- Quantify adjudication burden for each dimension under the core mapping.
- Repeat dimension summaries under the expanded mapping.
- Classify the claim depth supported by the current data for each dimension.
- Estimate dimension ranks for final unacceptable and adjudication rates.
- Test each dimension against the non-dimension response pool with false-discovery-rate control.
- Estimate model-category differences within each dimension.
- Evaluate whether adjudication identifies responses enriched for final unacceptable outcomes.
- Evaluate the D8 governance-extension pattern, including core vs expanded-only burden, rank inversion, model-level diffusion, and phenotype polarization.

6. Endpoints

Primary endpoint:

- Final unacceptable response, using the locked final unacceptable variable.

Key secondary endpoint:

- Third-round adjudication required, using the locked adjudication-required variable.

Additional descriptive endpoints:

- M2 consensus rate, reported as a descriptive agreement measure.
- Final failure phenotype among final unacceptable responses, using existing phenotype labels only.
- Model-level any-failure status for concentration and diffusion summaries.

7. Data Integrity Gates

Before producing inferential or descriptive tables, the script must verify:

- 636 unique prompt-model responses.
- 53 unique prompts.
- 12 unique models.
- 196 final unacceptable responses among unique prompt-model responses.
- 166 adjudication-required responses among unique prompt-model responses.
- No duplicate dimension, prompt, and model membership rows.
- All eight dimensions D1-D8 are present.
- Required columns are present: `dimension_id`, `set_membership`, `selected_prompt_index`, `prompt_model_id`, `model_code`, `model_category_collapsed`, `source_domain`, `final_unacceptable`, `adjudication_required`, `m2_consensus_rate_numeric`, and `final_failure_phenotype`.

If any gate fails, the analysis must stop and the failure must be recorded in the manifest.

8. Claim-Depth Classification

Claim depth will be assigned separately for the core mapping and expanded mapping using response support and event support.

Primary structured exploratory support:

- at least 144 dimension-response memberships;
- at least 12 prompts;
- at least 30 final unacceptable events;
- at least 20 adjudication-required events.

Secondary quantitative exploratory support:

- at least 96 dimension-response memberships;
- at least 8 prompts;
- at least 20 final unacceptable events or at least 20 adjudication-required events.

Limited qualitative/exploratory support:

- any dimension not meeting the secondary quantitative threshold.

Expanded mapping can support expanded-boundary interpretation but cannot upgrade a sparse core dimension into a primary core-axis claim. If core and expanded classifications differ, both must be reported.

9. Statistical Analysis

For each dimension and mapping layer, report:

- number of prompts, source domains, models, and dimension-response memberships;
- final unacceptable n/N, percentage, and Wilson 95% confidence interval;
- adjudication-required n/N, percentage, and Wilson 95% confidence interval;
- prompt-cluster bootstrap 95% interval for the final unacceptable rate;
- model-cluster bootstrap 95% interval for the final unacceptable rate;
- mean M2 consensus rate;
- rank among eight dimensions by final unacceptable rate and adjudication rate;
- claim-depth classification.

For dimension-vs-other contrasts:

- compare each dimension with the unique response pool not belonging to that dimension under the same mapping layer;
- report risk difference, risk ratio, odds ratio, Fisher exact two-sided p value, and Benjamini-Hochberg q value across D1-D8 separately by endpoint and mapping layer;
- interpret these tests as exploratory screening, with effect sizes prioritized over p values.

For model-category contrasts:

- compare medical-vertical and general-foundation models within each dimension and mapping layer;
- report n/N, risk difference, risk ratio, odds ratio, and Fisher exact p value at the response level;
- report exact model-label permutation p value using model-level dimension rates, preserving the 4 medical-vertical vs 8 general-foundation model split.

For adjudication as a risk signal:

- compare final unacceptable rates among adjudicated vs non-adjudicated responses for the overall corpus and for each dimension/mapping layer;
- report risk ratio, odds ratio, Fisher exact p value, sensitivity, specificity, positive likelihood ratio, negative likelihood ratio, and enrichment over the overall failure rate.

For D8 governance-extension analysis:

- compare D8 core with D8 expanded-only for final unacceptable and adjudication rates;
- report risk difference, risk ratio, odds ratio, Fisher exact p value, prompt-cluster bootstrap risk-difference interval, and model-cluster bootstrap risk-difference interval;
- compare D8 rank under core and expanded mapping;
- compare model-level any-failure status in D8 core vs D8 expanded-only using exact McNemar testing;
- report concentration metrics: models with any failure, top-3 failure share, Herfindahl-Hirschman index, effective number of contributing models, Gini coefficient, and normalized entropy;
- among D8 expanded final unacceptable responses, compare sycophancy/empathy misuse vs other phenotypes by model category, using Fisher exact p value and exact model-label permutation p value.

10. Multiplicity

Benjamini-Hochberg q values will be reported for the D1-D8 dimension-vs-other screening tests separately by endpoint and mapping layer. Other analyses are structured exploratory analyses and will not be converted into confirmatory claims.

11. Missing Data

No missing primary endpoint data are expected in the locked corpus. Missing failure phenotype values are expected for acceptable responses and will not be imputed. Phenotype analyses are restricted to final unacceptable responses with existing phenotype labels.

12. Reporting Rules

All results must report exact numerators and denominators. Percentages should be rounded to one decimal place. P values should be reported as exploratory and must not be used alone to justify claim strength.

The manuscript interpretation should use the following language hierarchy:

- Strongest allowed: "In this locked exploratory secondary analysis, dimension X had Y/Z responses with final unacceptable outcomes."
- Allowed: "The pattern suggests a candidate value-boundary signal that warrants independent validation."
- Not allowed: "The framework proves that dimension X causes failure" or "the model is unsafe in clinical deployment."

13. Planned Outputs

The analysis will produce:

- protocol/SAP Markdown and Word copies;
- data-integrity audit JSON;
- CSV tables for all prespecified analyses;
- an Excel workbook containing all tables;
- a concise Markdown and Word results report;
- an execution manifest with file paths and SHA-256 checksums.